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Articular Facets of the Human Spine

Quantitative Three-Dimensional Anatomy

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This study provides the quantitative three-dimensional surface anatomy of the articular facets for the entire human vertebral column based on a study of 276 vertebrae. Means and standard errors of the means for linear, angular, and area dimensions of the superior and inferior articular facets were measured for all vertebrae from C2 to L5. Facet orientations were described as angles with respect to the sagittal and transverse planes and also as card angles. The plane angles are similar to the angles seen on traditional radiographic views—radiographs and computed tomographic scans. The card angles, a new concept, are better at helping visualize the three-dimensional orientations of the facets. Excluding the superior C2 facet, the following minimum and maximum dimensions were found for the facets from C3 to L5: width = 9.6–16.3 mm; height = 10.2–18.4; surface area = 72.3–211.9 mm^2; interfacet width = 20.8–40.6; interfacet height = 12.2–33.0 mm; transverse plane angle = 41.0–86.0; sagittal plane angle = 67.4–154.8; X-card angle = 41.0–86.0; and Y-card angle = 5.8–66.1. The quantitative anatomy of the facets may improve the understanding of the spinal anatomy, help improve the clinical diagnosis and treatment, and provide the necessary data for constructing more realistic mathematical models of the spine. [Key words: spine, vertebrae, anatomy, vertebral dimensions, articular facets, zygapophyseal joints]

The shapes and sizes of human vertebrae are often artist's drawings based on visual inspection of cadavers,[15,16] with no quantitative geometric information provided. The quantification of vertebral articular facets into a geometric database has several applications. The orientation of the articular facets was hypothesized to be a cause of rotational coupling in the lumbar spine.[9] The role of the articular facets in intervertebral joint kinematics has been studied frequently.[1,2,10] The facet joints have been hypothesized to be a frequent source of low back pain.[4,21,29]

Another important use for quantified anatomy is the development of accurate and valid mathematical models of the human spine. Mathematical models depicting the spine have been used to study the mechanical response of the spine to external forces[30] and with accompanying spinal instrumentation.[14] Presently, the geometric data concerning vertebrae used in these spinal models is generally of much lower quality than the mathematical sophistication of the models. Because the accuracy of prediction by the models depends on the quality of the incorporated data, it is essential that high-quality data sets be available.

Despite the importance of the articular facets in a clinical setting and in the basic understanding of spine mechanics, little work has been done in quantifying their anatomy. Francis[12] documented the linear dimensions of the articular facets in the cervical spine and described their orientation. The facets at the thoracolumbar junction have been studied by Maqlmivaara.[20] Horwitz and Smith[17] and Skipor et al[31] studied facet orientations in the lumbar spine. More recently, Ahmed et al[3] described the facet geometry in the lumbar spine while studying its effect on axial torque response. Recent studies by Berry et al,[5] Scoles et al,[28] Gilad and Nissan,[13] Krag et al,[18,19] Zindrick et al[34] and Panjabi et al[24,25,26] have documented the anatomy of various vertebral elements in each region of the spine but did not focus on the articular facets.

The purpose of this article is to report quantitative three-dimensional surface geometry in the form of linear, areal, and angular dimensions of the articular facets of the spine from C2 to L5 vertebrae.

Materials and Methods

The data presented here contain the final stage in a series of anatomic studies. A detailed description of the methods is found in the initial document that described the surface anatomy of the thoracic vertebrae.[25] A brief description follows.

Twelve fresh autopsy spine specimens (axis to last lumbar vertebra) were obtained and radiographed to identify

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any significant abnormalities. The specimens were identified by age, sex, and height. The average age of the subjects was 46.3 years (range, 19–59), weight was 67.8 kg (range, 54–85), height was 167.8 cm (range, 157–178), and the male:female ratio was 8:4. Each specimen was prepared by completely removing all soft tissues from the vertebrae, thus disarticulating the 12 spines into 276 vertebrae. Two pointed threaded rods (2 mm diameter and at least 2.5 cm long) were secured into each vertebral body. The two ends of one rod constituted two marker points, while the third marker point was provided by one end of the second threaded rod. Anteroposterior and lateral radiographs were taken to identify the marker points in the vertebral body coordinate system with origin at the center of the upper end-plate. As an example, a lumbar vertebra with the coordinate system is seen from the anterior in Figure 1.

Linear dimensions, areas and angulations of various vertebral parts were studied, with the articular facets being the focus of the current study. A summary of all measured dimensions for the articular facets is listed in Table 1. The linear and area measurements are shown graphically in Figure 1, whereas the facet orientations, in terms of planar and card angles, are shown in Figure 2. Each articular facet surface was defined in space by a series of points spanning its surface. The number of points chosen was such that the geometry was well-defined by straight lines joining the adjacent points.

The three-dimensional coordinates of all the points with respect to a reference system defined by the three marker points were determined using a specially designed morphometer.[22] This instrument consisted of one linear and two rotational displacement transducers arranged such that their axes met at one point, thus establishing a spherical coordinate system (latitude, longitude, and radius). The location of any point, or series of points, with respect to the three marker points, was determined by placing the tip of the morphometer at the point(s) in question and at the three marker points, and recording the three-dimensional coordinates in a computer. In this manner, the morphometer was used to record the three-dimensional positions of all points which defined the vertebral parts listed in Table 1. In addition to the parts listed in Table 1, four points defining the left and right lower and upper edges of the posterior wall of the vertebral body were also digitized. These four points were used to define the vertebral body reference planes: frontal (XY), transverse (XZ), and sagittal (YZ), using a least squares technique. The position of the center of the upper end-plate was digitized as the origin for the vertebral body coordinate system, Figure 1. Four types of anatomic parameters were computed and are described later. Data were pooled from the 12 vertebrae at each level, and the mean and standard error of the mean for each dimension were computed.

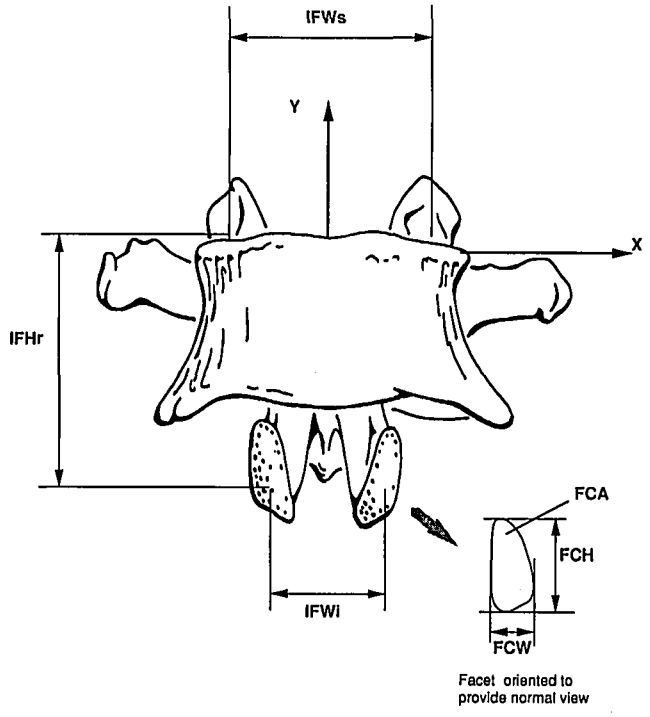


Figure 1. Linear and areal dimensions of the facets that were measured in the study, together with the nomenclature, are shown.

Facet Dimensions. Computer software was developed to calculate the various linear parameters from the three-dimensional digitized coordinates. The major and minor linear dimensions of each articular facet were calculated.

Facet Surface Areas. The projected areas of the digitized parts on to the frontal (XY), sagittal (YZ), and transverse (ZX) planes were computed by considering only two of the three coordinates of each point constituting the part. For example, to calculate frontal plane area, only the X and Y coordinates of the points were considered. The total area was computed as the euclidean sum (i.e., square root of the sum of the individual squares) of the three projected areas.

Interfacet Distance. The distances between the facets of each vertebra were calculated. Specifically, these interfacet

Table 1. Mnemonics for Each Facet Dimension

Mnemonics	Dimension
Linear	
FCH	Facet height
FCW	Facet width
IFH	Interfacet height
IFW	Interfacet width
Areal	
FCA	Facet area
Angular	
CAX	Card angle about X-axis
CAY	Card angle about Y-axis
S	Sagittal plane angle
T	Transverse plane angle
Suffixes	
i	Inferior
l	Left
r	Right
s	Superior

distances were the width between the centroids of the two superior facets and two inferior facets and the heights were between the left and right facets.

Facet Orientations. To calculate angulations, the surface of each facet was represented by a plane as shown in Figure 2A. Because each facet was characterized by more than three points, the equation of a best-fit plane to the facet surface was calculated using the least squares technique. To quantitatively represent the facet plane so obtained within the framework of the anatomic planes, we chose two alternatives.

Plane Angles. In the first method, the orientation of the facet plane was represented by two angles made with the anatomic planes: the sagittal (S) and transverse (T), as shown in Figure 2A. Note that the orientation of a three-dimensional plane is defined completely by two independent angles, such as S and T. The sign convention for the planar angles was defined as follows. The angle between the anatomic plane and the facet surface was deemed positive if oriented in the direction of the positive axis, which was perpendicular to the anatomic plane. For example, for the left facet, the sagittal plane angle was positive since it was in the direction of the positive X-axis.

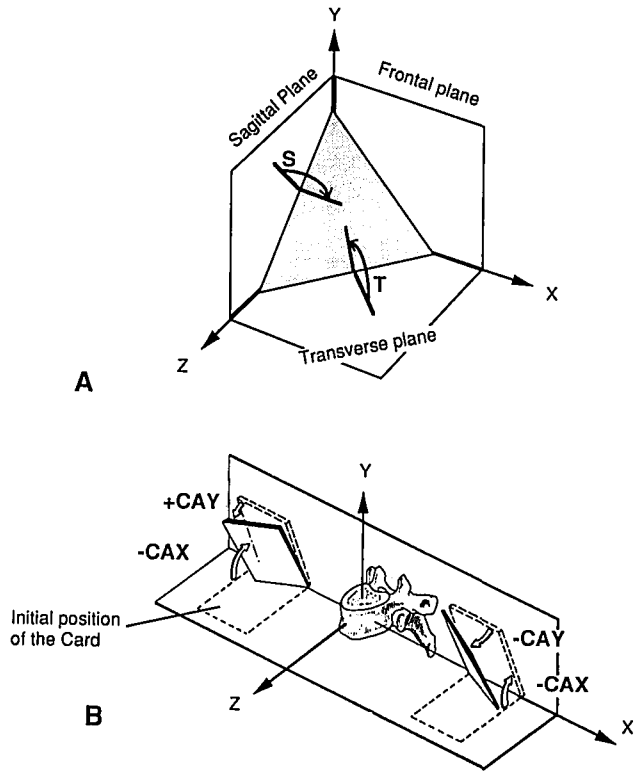


Figure 2. Facet surface was approximated by a plane. **A**, The orientation of the plane was defined by two angles made by the facet plane with the anatomic planes: sagittal (S) and transverse (T). **B**, Alternatively, the facet plane orientation was defined by two card angles: rotating a card, initially lying in the transverse plane, by angle CAX, followed by rotating the card around the Y-axis by angle CAY.

Card Angles. In the second alternative, a more intuitive representation is used. The orientation of the facet plane is described by two "card" angles CAX and CAY, as shown in Figure 2B. These card angles (CA) are rigid body rotations of a flat card, initially lying in the transverse plane, and then rotated, firstly about the X axis (CAX), and then about the Y axis (CAY). Note that in this representation, because of the sagittal plane symmetry, the CAX angles would be about equal for the left and right facets. Further, because of the sagittal plane mirror-image symmetry, the CAY angles for the left and right facets would be in the opposite directions, but would have about the same magnitude.

Accuracy of Measurements. The accuracies in computing linear, surface area, and angular parameters were estimated using an object with known values. A piece of Plexiglas (Rohm and Haas Co., Philadelphia, PA) with sides cut at 30,° 45,° and 60° was used. A sheet of millimeter graph paper was pasted on each of the cut surfaces and different shapes (triangles, rectangles, circles, etc.) were drawn on it. The three marker points, similar to those of a vertebra, were identified on the Plexiglas piece. Thus a pseudovertebra with surfaces of known inclinations and parts of known dimensions and areas was created. The parts were digitized with the same protocol and computer programs used for the vertebral parts. The linear, surface area, and angular parameters were computed and compared with the known values. The error was found to be less than 5% in all cases.

■ Results

A total of 12 C2 to L5 spines were studied. As described, the quantification of the articular facets was done in terms of linear dimensions, surface areas, interfacet distances, and facet orientations. The nomenclature for the measured parameters is given in Figures 1 and 2, and also in Table 1. The complete results, in the form of means and standard errors of the means, of the parameters measured for the vertebrae (C2 to L5) are listed in Tables 2–8, for the facet dimensions, facet surface areas, interfacet distances, and facet orientations (planar and card), respectively. The variations of the facet dimensions are best seen in Figures 3–7, and are described later.

Facet Dimensions

The superior facet widths and heights are shown in Figure 3A as the averages of the left and right values. The largest dimensions were found at C2, followed closely by those at the lower lumbar vertebrae, L4 and L5. The superior facets decreased in size significantly from C2 to C3. From C3 to T1, the facet dimensions were generally larger than those in the midthoracic region (i.e., T3–T8). The facet dimensions increased slightly toward the thoracolumbar junction region and then increased significantly in the lumbar region.

Table 2. Means (and Standard Errors of the Mean) of the Superior Articular Facet Linear Dimensions

Vertebra	FCW_{sr}	FCW_{sl}	FCH_{sr}	FCH_{sl}	FCW/FCH_{sr}	FCW/FCH_{sl}
C2	16.8 (0.3)	16.0 (0.5)	18.0 (0.4)	17.7 (0.6)	0.93	0.90
C3	11.4 (0.7)	10.8 (0.5)	12.5 (1.1)	10.6 (0.6)	0.91	1.02
C4	12.0 (0.6)	11.0 (1.0)	12.3 (1.2)	12.0 (1.1)	0.96	0.91
C5	12.3 (0.8)	11.8 (0.8)	12.0 (0.9)	11.2 (0.8)	1.04	1.05
C6	11.6 (0.5)	13.2 (0.7)	10.2 (0.6)	11.7 (0.6)	1.14	1.13
C7	12.3 (0.9)	12.8 (0.6)	12.1 (0.8)	10.7 (0.3)	1.02	1.20
T1	13.3 (0.7)	14.0 (0.9)	12.6 (0.6)	12.6 (1.2)	1.06	1.11
T2	11.7 (0.6)	12.1 (0.5)	12.2 (0.5)	13.0 (0.8)	0.96	0.94
T3	10.8 (0.5)	10.6 (0.3)	10.8 (0.4)	11.7 (0.5)	1.00	0.90
T4	10.5 (0.4)	10.3 (0.2)	11.0 (0.4)	11.5 (0.4)	0.95	0.90
T5	10.6 (0.3)	9.8 (0.3)	12.1 (0.5)	10.9 (0.5)	0.88	0.91
T6	9.6 (0.5)	10.1 (0.3)	11.7 (0.5)	11.8 (0.7)	0.83	0.86
T7	9.6 (0.5)	9.9 (0.4)	11.9 (0.7)	11.0 (0.4)	0.80	0.90
T8	10.0 (0.4)	10.3 (0.3)	11.5 (0.7)	11.7 (0.7)	0.87	0.89
T9	11.0 (0.4)	10.7 (0.3)	12.1 (0.7)	11.7 (0.7)	0.91	0.91
T10	11.5 (1.0)	12.9 (0.9)	12.0 (0.7)	12.4 (0.6)	0.96	1.03
T11	11.4 (0.4)	11.3 (0.5)	12.5 (0.5)	12.0 (0.6)	0.91	0.95
T12	11.2 (0.7)	10.5 (0.7)	13.4 (0.5)	12.5 (0.7)	0.84	0.84
L1	10.2 (0.6)	10.5 (0.6)	12.7 (0.3)	12.2 (0.7)	0.80	0.86
L2	11.1 (0.7)	11.4 (1.0)	14.6 (0.6)	14.6 (0.7)	0.76	0.78
L3	13.8 (0.9)	13.9 (0.7)	16.0 (0.8)	15.9 (0.8)	0.86	0.87
L4	14.1 (1.0)	15.3 (0.6)	16.1 (0.5)	17.3 (0.7)	0.88	0.89
L5	16.3 (0.7)	14.9 (0.6)	17.4 (0.7)	17.5 (0.7)	0.93	0.85

Refer to Table 1 for nomenclature.

Table 3. Means (and Standard Errors of the Mean) of the Inferior Articular Facet Linear Dimensions

Vertebra	FCW_{ir}	FCW_{il}	FCH_{ir}	FCH_{il}	FCW/FCH_{ir}	FCW/FCH_{il}
C2	11.4 (0.3)	10.4 (0.5)	13.0 (0.7)	11.8 (0.6)	0.88	0.88
C3	11.7 (0.3)	11.0 (0.7)	12.9 (1.0)	11.9 (1.2)	0.90	0.92
C4	12.2 (0.5)	11.1 (0.6)	13.0 (0.7)	11.5 (1.0)	0.94	0.97
C5	11.2 (0.6)	12.3 (0.8)	10.8 (0.5)	11.5 (0.5)	1.04	1.07
C6	12.7 (0.5)	13.1 (0.9)	12.3 (0.4)	12.2 (0.7)	1.03	1.07
C7	12.6 (0.7)	13.9 (0.7)	11.4 (0.5)	13.9 (0.9)	1.10	1.00
T1	11.9 (0.3)	13.2 (0.9)	12.5 (0.4)	13.3 (1.1)	0.96	0.99
T2	11.1 (0.6)	11.0 (0.5)	11.2 (0.5)	11.8 (0.4)	0.98	0.93
T3	9.9 (0.5)	10.4 (0.3)	11.0 (0.4)	10.7 (0.5)	0.90	0.97
T4	10.4 (0.4)	9.8 (0.3)	11.0 (0.4)	11.0 (0.4)	0.94	0.89
T5	9.7 (0.4)	9.7 (0.3)	10.9 (0.3)	11.2 (0.3)	0.89	0.87
T6	9.6 (0.4)	9.7 (0.3)	10.3 (0.4)	11.1 (0.3)	0.93	0.87
T7	9.8 (0.5)	10.3 (0.3)	10.6 (0.5)	11.0 (0.4)	0.92	0.94
T8	10.4 (0.4)	10.7 (0.3)	10.9 (0.5)	11.6 (0.2)	0.95	0.92
T9	11.9 (0.7)	12.6 (0.6)	12.3 (0.5)	12.5 (0.7)	0.97	1.01
T10	11.9 (0.5)	11.4 (0.3)	12.4 (0.3)	11.7 (0.5)	0.96	0.97
T11	11.4 (0.6)	10.9 (0.7)	13.3 (0.5)	12.5 (0.6)	0.86	0.87
T12	9.5 (0.6)	9.8 (0.7)	12.5 (0.5)	12.7 (0.7)	0.76	0.77
L1	12.4 (0.7)	10.7 (0.5)	15.3 (0.7)	15.7 (0.6)	0.81	0.68
L2	12.7 (0.7)	12.7 (0.8)	16.0 (0.6)	16.3 (0.6)	0.79	0.78
L3	13.8 (0.4)	13.4 (0.4)	15.7 (0.3)	16.4 (0.5)	0.88	0.82
L4	14.7 (0.5)	14.1 (0.6)	16.2 (0.7)	15.6 (0.7)	0.91	0.90
L5	15.6 (0.9)	16.1 (0.9)	18.4 (1.0)	17.3 (1.0)	0.85	0.93

Refer to Table 1 for nomenclature.

The average inferior facet widths and heights are shown in Figure 3B. The general trends were the same as described for the superior facets, that is, gradually increasing in size from C3 to C7, the smallest sizes in the midthoracic region, and again increasing in the lumbar region. One obvious difference between the inferior and superior facets was at C2 where the superior facets were particularly large while the inferior facets were effectively the same size as at the immediately subjacent vertebrae.

The ratios of facet width to height are shown in Figure 3C for the superior and inferior facets. The ratio increased from C2 to C7, decreased to T7, increased to T10 and decreased into the lumbar region.

Table 4. Means (and Standard Errors of the Mean) of the Superior and Inferior Articular Facet Surface Areas

Vertebra	FCA_{sr}	FCA_{sl}	FCA_{ir}	FCA_{il}
C2	211.3 (5.6)	205.1 (9.2)	94.6 (7.4)	85.0 (6.4)
C3	101.3 (15.2)	81.9 (8.5)	105.7 (10.5)	91.7 (19.3)
C4	104.2 (15.5)	95.6 (19.2)	105.3 (9.6)	85.7 (10.7)
C5	107.0 (15.8)	87.2 (12.5)	83.0 (7.1)	86.4 (9.4)
C6	79.5 (6.2)	94.6 (9.9)	97.8 (7.2)	94.4 (7.0)
C7	79.9 (5.5)	83.9 (6.5)	94.3 (3.5)	114.6 (9.1)
T1	102.9 (6.0)	114.8 (12.4)	84.5 (5.5)	94.4 (7.0)
T2	80.0 (6.9)	93.2 (10.0)	82.6 (4.9)	86.8 (6.5)
T3	79.7 (4.6)	85.8 (5.2)	73.3 (5.7)	78.9 (4.6)
T4	74.9 (4.7)	79.6 (4.0)	77.6 (3.4)	73.2 (3.9)
T5	85.6 (6.1)	73.5 (3.5)	72.3 (4.5)	74.6 (2.1)
T6	74.7 (5.4)	77.8 (4.8)	69.4 (4.1)	73.7 (3.1)
T7	75.8 (7.4)	74.6 (3.6)	69.6 (4.1)	76.7 (4.0)
T8	77.5 (4.8)	78.0 (5.3)	75.9 (2.0)	80.8 (2.6)
T9	82.1 (5.3)	83.3 (4.8)	89.5 (3.5)	99.0 (10.3)
T10	87.0 (7.5)	94.8 (13.5)	98.5 (5.3)	89.8 (6.2)
T11	91.9 (3.4)	87.7 (6.3)	100.3 (6.1)	92.4 (8.9)
T12	100.8 (9.1)	88.4 (9.2)	90.8 (9.1)	90.0 (7.2)
L1	96.7 (6.2)	99.2 (11.4)	127.9 (7.5)	125.9 (6.9)
L2	138.0 (11.4)	148.2 (12.8)	150.9 (9.8)	153.6 (10.1)
L3	170.3 (13.1)	164.4 (12.0)	159.8 (4.8)	167.9 (9.7)
L4	175.0 (7.0)	194.3 (11.0)	182.1 (15.6)	167.9 (11.8)
L5	211.9 (14.0)	199.1 (12.0)	197.3 (21.8)	182.7 (17.2)

Refer to Table 1 for nomenclature.

Table 5. Means (and Standard Errors of the Mean) of the Interfacet Distances

Vertebra	IFW_s	IFW_i	IFH_r	IFH_l	IFW_s/IFW_i	IFW/IFH
C2	30.3 (0.6)	35.9 (0.8)	20.1 (0.9)	20.2 (0.8)	0.84	1.64
C3	35.9 (0.6)	37.8 (0.9)	12.2 (0.4)	13.0 (0.5)	0.95	2.93
C4	37.6 (0.8)	38.0 (0.8)	15.8 (3.6)	15.9 (3.3)	0.99	2.38
C5	38.3 (0.9)	40.5 (0.9)	13.3 (0.7)	12.7 (0.6)	0.94	3.04
C6	39.9 (0.7)	38.8 (0.8)	13.9 (1.1)	13.5 (0.8)	1.03	2.88
C7	38.6 (0.8)	33.6 (0.8)	17.6 (0.7)	17.0 (0.7)	1.15	2.09
T1	33.2 (1.0)	28.2 (0.6)	18.0 (0.6)	18.6 (0.7)	1.18	1.68
T2	28.2 (0.5)	24.3 (0.6)	21.2 (0.6)	20.9 (0.8)	1.16	1.25
T3	24.6 (0.7)	22.3 (0.7)	21.5 (0.7)	21.1 (0.8)	1.10	1.10
T4	22.1 (0.8)	21.1 (0.7)	20.9 (0.6)	20.1 (0.6)	1.05	1.04
T5	20.8 (0.8)	20.8 (0.8)	21.2 (0.8)	21.1 (0.7)	1.00	0.98
T6	20.9 (0.9)	21.4 (0.7)	21.8 (0.7)	22.0 (0.5)	0.97	0.97
T7	21.0 (0.7)	22.3 (0.9)	24.0 (0.8)	23.6 (0.7)	0.94	0.91
T8	21.8 (0.9)	23.4 (1.0)	24.0 (0.5)	25.2 (0.4)	0.93	0.92
T9	23.0 (1.0)	24.6 (0.9)	25.7 (0.8)	25.9 (0.8)	0.93	0.92
T10	24.2 (0.9)	24.6 (1.1)	26.1 (0.5)	25.8 (0.7)	0.99	0.94
T11	25.2 (1.2)	24.6 (0.7)	29.9 (1.1)	29.8 (1.0)	1.02	0.83
T12	25.9 (0.8)	24.9 (0.4)	31.6 (1.2)	31.1 (1.5)	1.04	0.81
L1	26.2 (0.6)	24.8 (0.8)	32.5 (0.5)	31.8 (1.0)	1.06	0.79
L2	26.4 (0.9)	26.6 (0.9)	33.0 (0.7)	32.4 (0.6)	0.99	0.81
L3	28.6 (1.0)	29.1 (1.5)	32.4 (0.9)	31.6 (0.7)	0.98	0.90
L4	31.4 (1.6)	34.8 (1.7)	28.5 (1.0)	28.3 (1.3)	0.90	1.16
L5	35.0 (2.1)	40.6 (2.0)	26.1 (1.8)	26.4 (1.5)	0.86	1.44

Refer to Table 1 for nomenclature.

Note that the trends were the same for both the superior and inferior facets.

Facet Surface Areas

The surface areas of the articular facets are shown in Figure 4. These curves support the data presented earlier on the linear dimensions of width and depth. The superior facet areas were largest at C2 and in the lumbar region. From C3 to T12, the areas were remarkably constant. The inferior facet areas mimicked the superior areas except at C2 where the inferior facet area was the same as at the lower cervical verte-

Table 6. Means (and Standard Errors of the Mean) of the Superior Articular Facet Orientations with Respect to the Anatomic Planes: Transverse and Sagittal

Vertebra	T_{sr}	S_{sr}	T_{sl}	S_{sl}
C2	40.7 (4.7)	-114.1 (2.9)	33.5 (2.1)	118.5 (1.7)
C3	48.8 (3.1)	-76.8 (3.9)	55.2 (3.2)	71.6 (5.3)
C4	47.0 (4.6)	-83.9 (1.3)	49.7 (4.8)	85.2 (1.3)
C5	45.8 (2.9)	-83.4 (1.3)	45.3 (2.8)	83.7 (1.1)
C6	47.2 (2.8)	-81.2 (3.1)	48.0 (5.4)	76.8 (1.7)
C7	59.4 (4.5)	-80.2 (2.6)	51.6 (5.2)	85.5 (1.5)
T1	62.2 (2.6)	-86.4 (0.9)	57.7 (3.8)	85.3 (1.1)
T2	66.5 (1.9)	-75.5 (1.7)	66.6 (1.8)	74.4 (2.3)
T3	69.4 (2.2)	-74.7 (1.4)	69.4 (1.2)	74.3 (1.9)
T4	72.8 (1.6)	-77.2 (1.6)	73.2 (1.2)	74.6 (1.2)
T5	73.2 (2.2)	-73.4 (1.1)	74.5 (1.7)	75.3 (1.3)
T6	73.5 (1.5)	-72.5 (1.1)	74.6 (1.3)	72.9 (1.1)
T7	74.7 (0.8)	-73.5 (1.3)	72.7 (1.1)	74.1 (1.6)
T8	74.5 (1.2)	-72.2 (1.3)	73.0 (1.3)	74.4 (1.8)
T9	77.7 (0.4)	-74.0 (1.2)	78.2 (1.3)	73.3 (1.2)
T10	78.1 (1.1)	-76.2 (1.7)	78.1 (2.3)	74.0 (1.6)
T11	75.6 (1.1)	-77.0 (1.8)	76.0 (1.7)	71.6 (2.4)
T12	77.1 (2.1)	-72.5 (5.5)	78.2 (2.6)	67.4 (5.1)
L1	82.9 (1.4)	-139.1 (8.9)	81.3 (2.7)	134.2 (8.2)
L2	85.7 (0.9)	-135.4 (9.2)	83.8 (1.3)	139.7 (9.0)
L3	81.9 (1.6)	-131.5 (9.0)	84.1 (1.5)	133.1 (5.5)
L4	81.2 (1.6)	-120.8 (3.7)	81.3 (2.5)	123.6 (3.1)
L5	86.0 (1.0)	-117.6 (4.1)	85.1 (1.2)	119.8 (4.5)

Refer to Table 1 for nomenclature.

Table 7. Means (and Standard Errors of the Mean) of the Inferior Articular Facet Orientations with Respect to the Anatomic Planes: Transverse and Sagittal

Vertebra	T_{ir}	S_{ir}	T_{il}	S_{il}
C2	31.4 (5.2)	-74.3 (3.4)	36.1 (4.9)	70.8 (3.5)
C3	44.2 (6.5)	-75.2 (3.1)	41.0 (4.1)	79.4 (2.4)
C4	44.9 (5.7)	-74.3 (2.8)	47.7 (6.3)	75.0 (5.0)
C5	49.1 (4.7)	-75.1 (3.6)	54.3 (3.1)	72.0 (4.5)
C6	55.0 (4.0)	-81.6 (2.3)	54.7 (4.6)	83.1 (2.1)
C7	49.8 (2.9)	-82.4 (1.4)	49.1 (2.7)	80.1 (3.6)
T1	66.3 (2.6)	-75.5 (2.2)	62.2 (5.3)	75.4 (2.8)
T2	69.6 (2.3)	-75.8 (1.1)	70.0 (2.4)	76.5 (1.8)
T3	73.8 (1.8)	-75.1 (1.4)	75.4 (2.2)	74.9 (1.2)
T4	72.7 (3.0)	-75.3 (1.0)	74.4 (2.6)	73.7 (1.5)
T5	76.8 (2.8)	-72.2 (1.3)	76.4 (2.7)	73.6 (1.7)
T6	77.8 (2.2)	-73.5 (1.7)	76.7 (2.7)	74.7 (1.3)
T7	77.6 (2.1)	-73.8 (1.2)	77.0 (1.5)	74.6 (1.8)
T8	74.3 (2.0)	-75.0 (1.3)	77.1 (1.6)	74.7 (1.4)
T9	77.9 (2.1)	-75.7 (1.6)	77.5 (2.4)	75.4 (1.4)
T10	77.9 (1.7)	-76.8 (2.2)	76.7 (2.4)	74.0 (2.0)
T11	75.5 (2.1)	-72.2 (5.3)	76.1 (2.1)	70.5 (4.7)
T12	75.4 (1.9)	-139.7 (9.5)	78.4 (3.0)	138.8 (8.8)
L1	81.4 (1.6)	-152.1 (5.8)	81.8 (1.9)	154.8 (6.3)
L2	82.8 (2.5)	-142.5 (5.1)	78.7 (4.0)	136.7 (8.1)
L3	75.6 (4.1)	-130.6 (6.3)	77.8 (3.3)	123.3 (3.3)
L4	70.5 (6.7)	-112.7 (7.1)	80.3 (1.2)	114.7 (4.9)
L5	71.0 (3.7)	-127.6 (7.2)	81.6 (1.2)	125.1 (2.9)

Refer to Table 1 for nomenclature.

brae. Using the measured width and height of each facet, the facet area was estimated by assuming its shape was elliptical (i.e., articular facet surface area = π [facet width × facet height]/4). The results of the elliptical approximations are shown also in Figure 4.

Interfacet Distances

Interfacet distances between pairs of facets were calculated and are shown in Figure 5A. The lateral distance between superior or inferior facets was termed the interfacet width whereas the vertical distance be-

Table 8. Average Card Angles for the Superior and Inferior Articular Facets

Vertebra	CAX_{sr}	CAY_{sr}	CAX_{sl}	CAY_{sl}	CAX_{ir}	CAY_{ir}	CAX_{il}	CAY_{il}
C2	-40.7	38.9	-33.5	-59.9	-31.4	-31.3	-36.1	33.9
C3	-48.8	-17.6	-55.2	22.6	-44.1	-21.5	-41.0	16.3
C4	-47.0	-8.4	-49.7	6.4	-44.9	22.5	-47.7	-20.6
C5	-45.8	9.2	-45.3	-8.9	-49.1	19.9	-54.3	-22.3
C6	-47.2	12.1	-48.0	-17.9	-55.0	10.3	-54.7	-8.5
C7	-59.4	11.4	-51.6	-5.8	-49.8	9.9	-49.1	-13.1
T1	-62.2	4.1	-47.7	-5.6	-66.3	15.9	-62.2	-16.5
T2	-66.5	15.8	-66.6	-17.0	-69.6	15.1	-70.0	-14.4
T3	-69.4	16.4	-69.4	-16.8	-73.8	15.5	-75.4	-15.6
T4	-72.8	13.4	-73.2	-16.1	-72.7	15.4	-74.4	-16.9
T5	-73.2	17.3	-74.5	-15.3	-76.8	18.4	-76.4	-16.9
T6	-73.5	18.2	-74.6	-17.8	-77.8	18.4	-76.7	-15.8
T7	-74.7	17.2	-72.7	-16.7	-77.5	16.6	-77.0	-15.8
T8	-74.5	18.5	-73.0	-16.3	-74.3	15.6	-77.1	-15.7
T9	-77.7	16.4	-78.2	-17.1	-77.9	14.7	-77.5	-15.0
T10	-78.1	14.1	-78.1	-16.4	-77.9	13.5	-76.7	-16.5
T11	-75.6	13.5	-76.0	-19.0	-75.5	18.4	-76.1	-20.1
T12	-77.1	18.0	-78.2	-23.1	-75.4	-52.0	-78.4	50.2
L1	-82.9	-50.5	-81.3	44.9	-81.3	-63.4	-81.7	66.1
L2	-85.7	-45.6	-83.8	50.1	-82.8	-53.1	-78.7	47.9
L3	-81.9	-42.0	-84.1	43.4	-75.6	-42.2	-77.8	34.2
L4	-81.2	-31.2	-81.3	34.0	-70.5	-24.2	-80.3	25.1
L5	-86.0	-27.7	-85.1	29.9	-71.0	-40.2	-81.6	35.5

Refer to Table 1 for nomenclature.

tween left or right facets was termed the interfacet height. The interfacet width was largest in the cervical region and decreased sharply in the upper thoracic region to T5. From T5, the width rose gradually to L1 where it increased more sharply to L5. The average of the left and right interfacet heights are shown in Figure 5A. The interfacet height was lowest in the cervical region, increased gradually through the thoracic and upper lumbar regions and decreased at L4 and L5.

Ratios of the superior to inferior interfacet widths are shown in Figure 5B. From the middle to lower cervical region, this ratio increased above 1.0, implying that the superior interfacet width became larger than the inferior width. From T1 to T9, the ratio decreased thus implying that the inferior width was becoming larger than the inferior. Similarly, from T9 to L1 the ratio increased and decreased again from L1 to L5.

The ratio of the interfacet width (averaged superior and inferior) to the interfacet height (averaged left and right) was calculated and is shown in Figure 5C. Note that the ratio had a magnitude of between 1.5 and 3 in the cervical region and decreased to a magnitude of approximately 1 in the thoracic and lumbar regions. At the lowest lumbar vertebrae, the ratio began to increase.

Angular Dimensions

Plane Angles: The orientations of the articular facets with the sagittal and transverse planes are shown in Figures 6A and 6B, respectively.

The superior facet angle decreased significantly from 116° at C2 to 75° at C3. Continuing down the cervical spine, the average of the left and right sagittal plane angles increased from 75° at C3 to 80° at C7, as shown in Figure 6A. In the thoracic spine from T2 to T11, these angles remained constant at approximately 70–75°. At the thoracolumbar junction, there was a dramatic change in the facet angles with respect to the sagittal plane. This change occurred at the inferior facets of T12 and the superior facets of L1. The superior facet angle changed from approximately 70° at T12 to 137° at L1. With progression down the lumbar spine, the sagittal angle decreased from approximately 137° at L1 to 118° at L5.

In Figure 6B, the angle of the facets with the transverse plane are shown. From C2 to T4 the angle increased gradually from approximately 35° to 75°. From T4 to T12, this angle remained relatively constant. In the lumbar spine, the transverse plane angle was slightly larger, approximately 80°.

Although not presented here, the angle with the frontal plane could be calculated from the transverse and sagittal plane angles. Effectively, the frontal plane angles are complementary to the transverse plane angles in the cervical and thoracic regions and to the sagittal plane angles in the lumbar region.

Card Angles: The average card angles for each vertebra from C2 to L5 are plotted in Figures 7A and 7B for the superior and inferior facets, respectively. These angles were calculated directly from the average planar angles shown in Tables 5A and 5B. Within

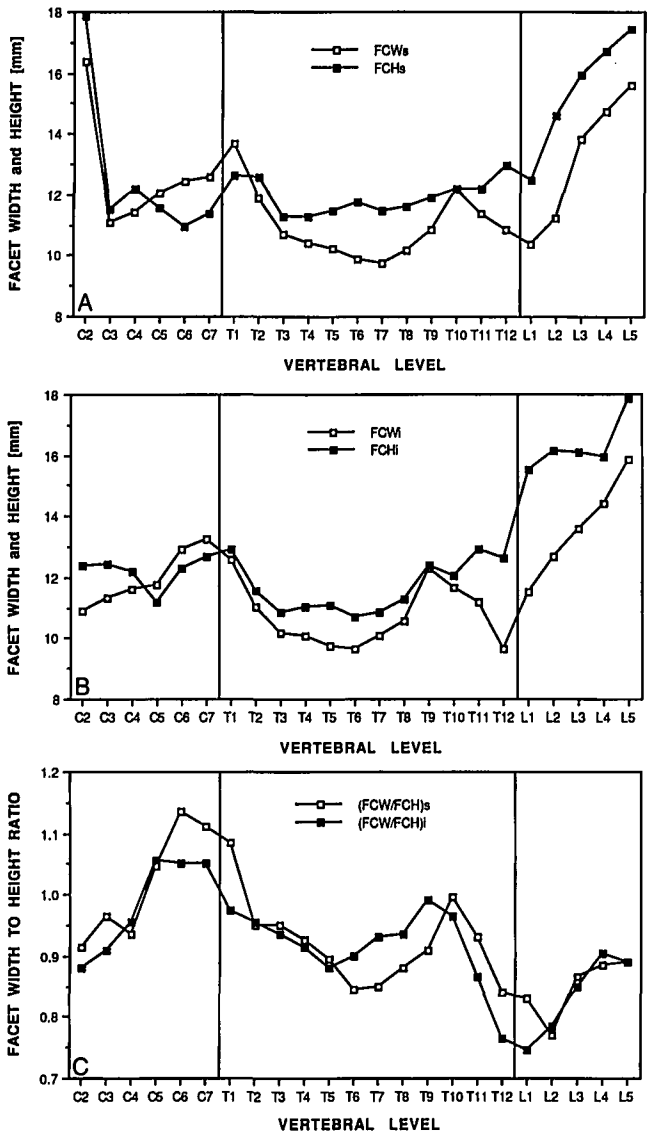


Figure 3. Width and height of facet as function of vertebral level. **A**, Superior facet. **B**, Inferior facet. **C**, Ratio of width to height for superior and inferior facets.

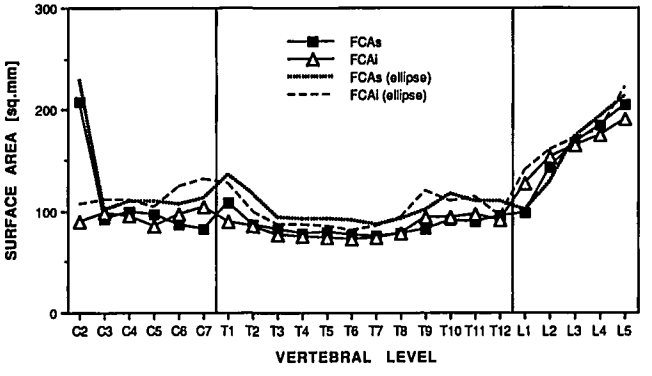


Figure 4. Facet surface areas of the superior and inferior facets as functions of vertebral level. Also shown are the approximations considering the facet as an ellipse with facet width and height as given in Tables 2A and 2B.

each figure, the card angles are shown for both the right and left facets. Recall that these card angles are easily visualized by rotations of a flat card initially lying in the transverse plane. The first rotation (CAX) is about the X-axis, which is followed by a second rotation (CAY) about the Y-axis. The signs of these rotations are as defined by the coordinate system. As seen in Figure 7A for the superior facets, the X-axis card angle CAX changed more or less in a linear fashion from about -41° at C2 to -86° at L5 with no dramatic changes in between. Conversely, the Y-axis card angle CAY changed at several locations along the spine. It completely reversed its direction at three levels: C2–C3; C4–C5; and T12-L1.

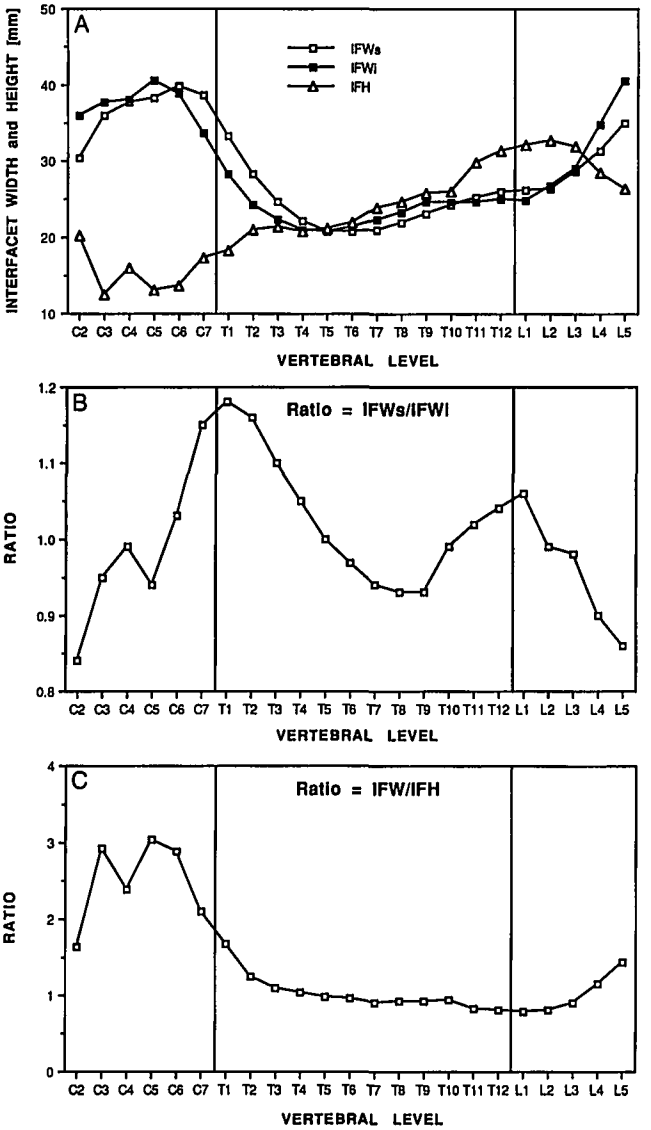


Figure 5. Interfacet width and height as functions of vertebral level. **A**, Widths of superior and inferior facets and average height between the superior and inferior facets. **B**, Ratio of superior to inferior interfacet widths. **C**, Ratio of average of left and right interfacet width to average of left and right interfacet height.

As an example, we will describe the right superior facet orientations here. The left facet angles are mirror images. Starting from +38.9° at C2, it changes to -17.6° at C3, and +9.2° at C5. It remains positive with a slight overall increase from C5 to T12, but it has a dip at T1 where it is +4.1°. Between T12 and L1, the superior facet shows dramatic change in its orientation as represented by the Y-axis card angle CAY, that is, the respective CAY values are +18.0° and -50.5°. Then onward, there is a continuous decrease to -27.7° at L5.

Card angles for the inferior facets are depicted in Figure 7B. As the inferior facets articulate with the superior facets of the vertebra below, the inferior facet angles are about the same as the superior facets, but are shifted one level upward. For example, the average CAX and CAY values for the right inferior facet of C6 are -55.0° and 10.3°, respectively; whereas the corresponding values for the superior facet of C7 are -59.4° and 11.4°. These inferior–superior facet angle relations follow throughout the spine. The exceptions are the superior facets of C2 and inferior facets of L5. The former articulates with the occiput, whereas the latter is part of the sacrum.

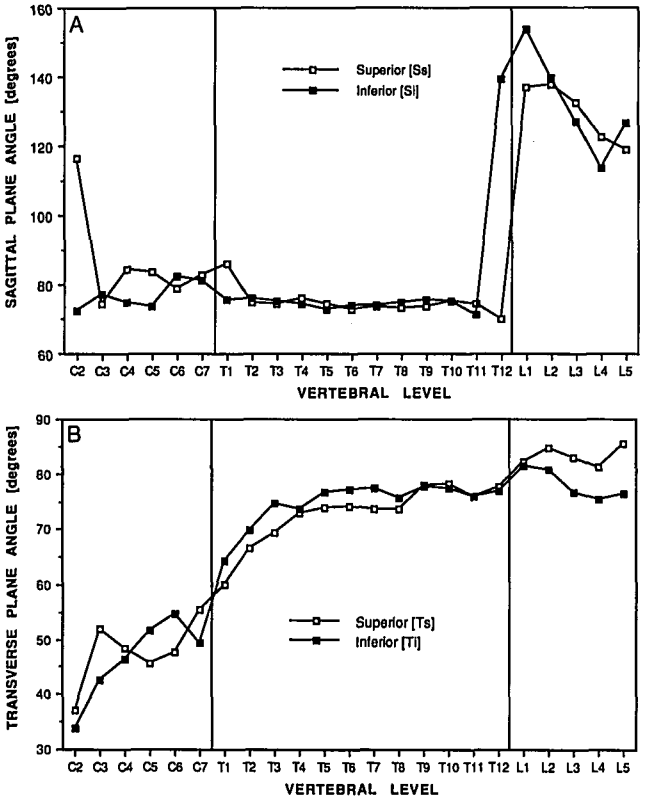


Figure 6. Angles made by the facet plane with the two anatomic planes as functions of vertebral level. **A**, Sagittal plane. **B**, Transverse plane. The average of the absolute values for both the left and right inferior and superior facet orientations are shown.

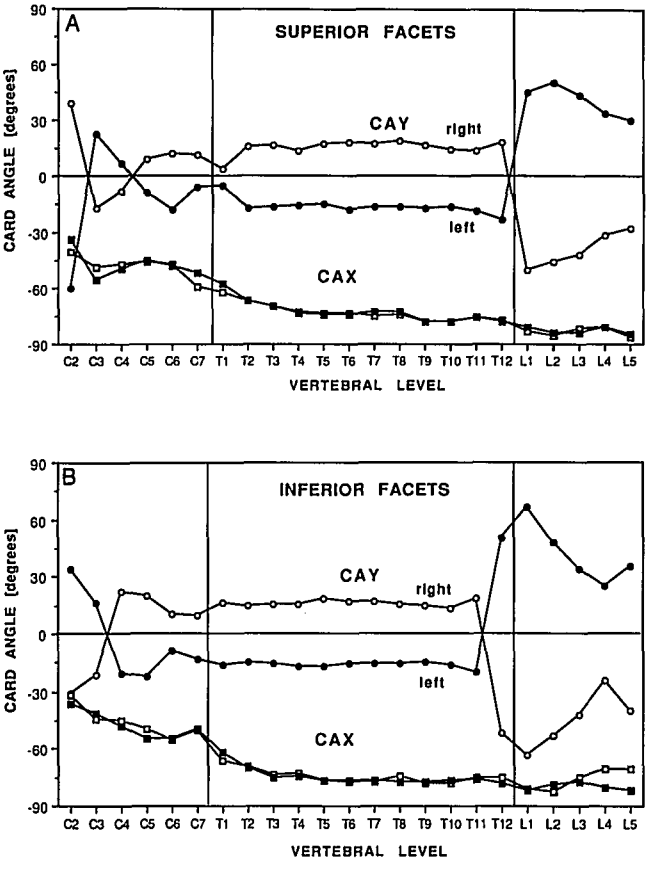


Figure 7. Card angles CAX and CAY for the right and left facets are shown as functions of vertebral level. **A**, Superior facets. **B**, Inferior facets.

The card angles quantify facet orientations in an alternative and, we believe, a more instructive manner. To exemplify the intuitive nature and easy visualization of the three-dimensional orientation of the facets that card angles provide, a representative facet from each region of the spine is shown in Figure 8. The values of the two card angles for the superior right facets of vertebrae C5, T6 and L3 are shown on the left side of the figure.

■ Discussion

To our knowledge, the current study is unique in that it provides, for the first time, a comprehensive quantitative three-dimensional surface anatomy of the articular facets of the entire vertebral column. Problems of spinal instability and spinal trauma may benefit from this quantified anatomic database, through the use of more precise dimensions of the facets and the development of more anatomically accurate mathematical models.

In the current study, the surface of the articular facets was approximated as a plane. However, it is known that the facets are not planar but are curved, especially in the lumbar region. To precisely describe the curved surfaces for all vertebrae requires additional work above and be-

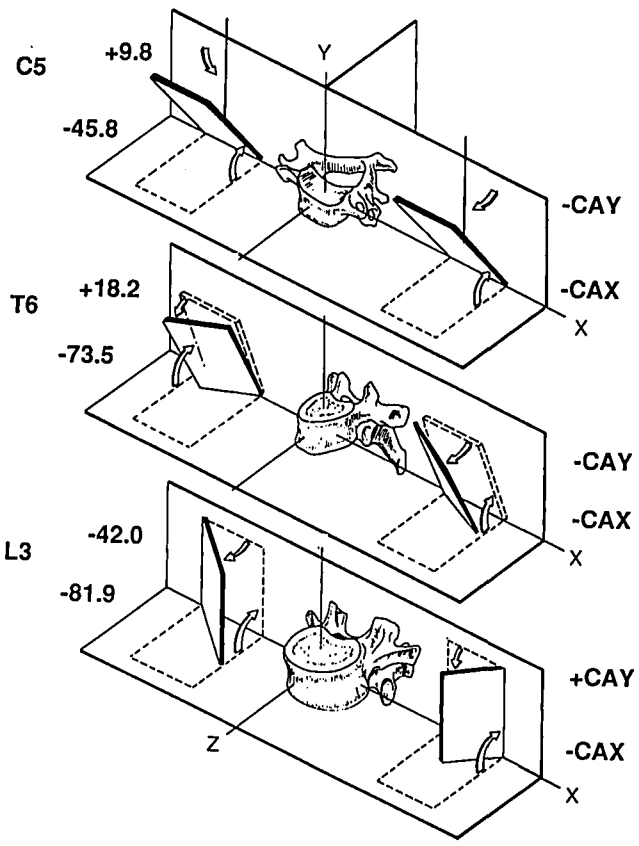


Figure 8. Facet plane orientation is quantified by two card angles, rotation CAX about the X-axis, followed by rotation CAY about the Y-axis, as shown on the right. As examples of variations in the three regions of the spine, the average card angles for right superior facets of C5, T6, and L3 are shown on the left. Note the gradual increase in the CAX from C5 to L3, and a sudden reversal of direction in the lumbar region for the CAY.

yond that presented here. For the purposes of the current study, we believed a planar approximation for the articular facets was warranted and most easily understood.

The concept of representing the facet angles by the use of a card has been used here for the first time to quantitate spatial orientations of the facets, although it was introduced earlier for easy qualitative visualization.[27] There are several other methods of representing spatial orientations of a plane, for example, direction cosines of a unit vector normal to the plane surface and angles made by the plane with the three planes of a coordinate system. However, we believe the card angle representation is unique: it is the easiest method to visualize the three-dimensional orientation of facets and is also extremely powerful in delineating the anatomic changes that differentiate the facet angles from C2 to L5.

The Regions of the Spine

From the data presented here, several geometric parameters about the articular facets may be identified as characterizing the different regions of the spine. These are outlined in the following paragraphs for each of the cervical, thoracic, and lumbar regions.

Cervical: The cervical vertebrae, using linear dimensions, are best distinguished from the thoracic and lumbar vertebrae by the ratio of the interfacet width to height. If this ratio was above 1.5, then the vertebra was almost certainly from the cervical region. Within the cervical region, the largest facet was the superior facet of vertebra C2. The angular characteristics that single out the cervical vertebrae are the relatively small angles made with the transverse plane (i.e., 30–50°), and large angles made with the sagittal plane. The card angles help separate the cervical spine from other regions, but also help divide the cervical spine into three regions: C2, C3–C4, and C5–C7. From one region to another there is a change in the sign of the CAY angle. This division is supported by the facet width to height ratio. This ratio (average of superior and inferior facets) is the smallest for C2, is about the same for C3–C4, and increases to a higher value for C5–C7.

Thoracic: The thoracic vertebrae are best differentiated from the cervical and lumbar vertebrae by their uniformity in most linear and areal dimensions throughout this region. The transverse plane angle of the facets is part of the uniform increase from 35° at C2 to 80° at L5. The CAX angle decreased from C2 to L5 nearly linearly; thus it may be used to distinguish the thoracic vertebrae from other regions. To individually differentiate the thoracic vertebrae was more difficult, since the thoracic articular facets exhibited little individual variation from T1 to T12. The CAY angle was approximately 16° throughout the thoracic region, similar to that of the middle and lower cervical spine regions, but very different from the lumbar spine region. The most significant difference between the individual thoracic vertebrae was the interfacet superior to inferior width ratio which decreased from 1.15 at T1 to 0.95 at T9. Vertebra T12 is easily identified by angular parameters, the abrupt change in the sagittal plane angles and CAY card angles of its superior to inferior facets.

Lumbar: The lumbar vertebrae can be differentiated from the cervical and thoracic vertebrae using several measured parameters. Certainly, one of the most characteristic parameters is the angle made with the sagittal plane. The facets of the lumbar vertebrae are oriented between 120° and 150° from sagittal plane whereas the cervical and thoracic vertebrae are angled from 60–90° with the same plane. The card angles also clearly differentiate the lumbar facets from the other two spinal regions and within the lumbar region itself. The CAX angle is highest, about

Table 9. Comparison with Previous Studies: Linear Dimensions, Facet and Interfacet Widths and Heights

Parameter	Level	Present Study	Francis (1955)	Cotterill (1986)
FCH_s	C2	17.8	18.8	—
	C3	11.5	11.3	—
	C4	12.0	11.5	—
	C5	11.3	11.1	—
	C6	10.8	10.6	—
	C7	11.0	10.2	—
FCW_s	C2	16.4	17.5	—
	C3	11.0	12.0	—
	C4	11.3	12.4	—
	C5	12.0	12.7	—
	C6	12.3	13.0	—
	C7	13.4	13.6	—
IFW_s	T6	20.9	—	19.0
	T12	25.9	—	22.5
	L3	28.6	—	26.6
IFH	T6	21.9	—	20.4
	T12	31.4	—	29.4
	L3	32.0	—	29.9

90°, in this region, but is about the same for all lumbar vertebrae. Conversely, The CAY first changes signs from T12 to L1 (superior right facet), and then decreases in magnitude from about 50° at L1 to 28° at L5. Other parameters characteristic of the lumbar vertebrae are relatively large facet widths, heights, and surface areas. To differentiate the lumbar vertebrae from one another, a combination of parameters may be used.

Fawcett[11] suggested that the geometric shapes formed by connecting the outer tips of the articular facets in the frontal plane can be used to differentiate the different lumbar vertebrae. In this scheme, the two upper lumbar vertebrae are identified by the larger interfacet width of the superior compared to the inferior facets. For the lowest three vertebrae, the shape changes from an upright rectangle at L3 to a square at L4 to a horizontal rectangle at L5. Our data are consistent with the findings of Fawcett. At L1, the superior interfacet width is greater than the inferior interfacet width. The ratio of the superior to inferior interfacet widths decreased from L1 to L5 (see Figure 5B). According to Fawcett, this ratio should decrease from L1 to L3, and remain constant to L5. For the lower lumbar vertebrae, Fawcett would predict the ratio of the interfacet width to height to be less than one at L3, equal to one at L4, and be greater than one at L5. Our measured interfacet width to height ratios were 0.9 at L3, 1.16 at L4, and 1.44 at L5, therefore supporting the classification scheme proposed by Fawcett.

Previous Studies

Several quantitative anatomy studies have been completed for the articular facets in particular regions of the spine: cervical[12] and thoracolumbar.[3,5,8,17,20,28,31,33] However, none are as comprehensive as the current study. Tables 9 and 10 provide comparisons among the various linear and angular parameters measured in the current study and some of the earlier work.

Linear geometric parameters of facet height and width in the cervical spine compare favorably with those reported previously by Francis.[12] Similarly, our interfacet distance compares favorably with the values reported by Cotterill et al[8] in the thoracolumbar region. There has been considerably more interest in the orientation of the articular facets. Several studies documented the lumbar facet angulation with the sagittal plane, using computed tomographic scan[3,33] and gross

Table 10. Comparison with Previous Studies: Angular Dimensions and Sagittal Plane Angle of Superior Vertebrae

Parameter	Level	Present Study	Malmivaara (1987)	Ahmed (1990)	Van Schaik (1985)	Horwitz & Smith (1940)
S_s	T11	75	73	—	—	—
	T12	70	98	—	—	—
	L1	137	117	—	—	155
	L2	137	—	—	—	150
	L3	132	—	151	—	135
	L4	122	—	—	143	135
	L5	118	—	129	132	135
	S1	—	—	—	127	—

dissection. [17,20,31] The computed tomographic measurements of Ahmed et al[3] and Van Schaik et al[33] were consistent and somewhat less than the sagittal plane angles reported here (Table 10). However, all five studies noted the increase in the sagittal plane angle with progression from L1 to L5. This trend was observed qualitatively by Taylor and Twomey,[32] Skipor et al[31] reported an average of 18° forward tilt of the lumbar inferior facets with a range of -28° to 53°. We observed a range from 19.5° to 8.7°.

The sagittal plane angles reported here are true angles, that is, angles between the facet and sagittal planes, whereas the angles reported by previous authors are angles between the two intersection lines of the facet and sagittal planes made with the transverse plane (computed tomographic slice). This could account for the differences in the angle comparisons of Table 10. For the computed tomographic angles with respect to the sagittal plane to be accurate, the transverse plane angles must be approximately 90° (i.e., lumbar region). Deviations from 90° would result in the computed tomographic angle being less than the true angle. In a similar manner, the facet angles with respect to the transverse plane can be estimated from lateral radiographs with accuracy only if the sagittal plane angles are approximately 90° (i.e., cervical and thoracic regions).

Geometrical Shape Approximations

As an aid to mathematical modeling of the spine, it is useful to be able to approximate various vertebral components by simple geometric shapes. In this study, the surface areas of the articular facets were approximated as ellipses based on their measured widths and heights. For both the superior and inferior articular facets, the elliptical approximation slightly overestimated the measured area.

■ Summary

A comprehensive data set, based on the three-dimensional study of 276 vertebrae, has been presented providing quantitative anatomy of articular facets for the entire spinal column. Such information is essential for the understanding of the facet function and the development of advanced and valid mathematical models of the spine. The latter may be used to predict spinal behavior, for example, effects of surgery and injury mechanisms.

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